

7.4 Sufficiency, completeness and minimum variance

The Gauss–Markov theorem compares least squares only with *linear* unbiased estimators, and Example (Uniform errors and the midrange) showed that the restriction matters: with uniform errors a nonlinear estimator wins easily. This section shows that under normality the restriction can be dropped. In the normal linear model, $\lambda^\top \hat{\beta}$ has the smallest variance among *all* unbiased estimators of an estimable $\lambda^\top \beta$, and s^2 has the smallest variance among all unbiased estimators of σ^2 . The argument is the classical one through sufficiency and completeness. It also produces best unbiased estimators of nonlinear functions, such as σ itself. At the end we compare the variances with the Cramér–Rao lower bound, which least squares attains for β and s^2 does not attain for σ^2 .

Two facts from mathematical statistics are quoted without proof: the factorization criterion for sufficiency and the completeness of full-rank exponential families. Both are proved in standard texts such as Lehmann and Casella (1998) and Lehmann and Romano (2005). Everything built on them is proved here.

7.4.1 Sufficiency

Definition 7.4.1 — Sufficient and complete statistics

Let $\mathbf{Y}$ have a distribution P_θ from a family indexed by $\theta \in \Theta$, and let $\mathbf{T} = \mathbf{T}(\mathbf{Y})$ be a statistic.

- (a) $\mathbf{T}$ is **sufficient** for θ if the conditional law of $\mathbf{Y}$ given $\mathbf{T}$ is the same for every θ .
- (b) $\mathbf{T}$ is **complete** if, for every function h with $E_\theta |h(\mathbf{T})| < \infty$ and $E_\theta h(\mathbf{T}) = 0$ for all $\theta \in \Theta$, we have $\Pr_\theta\{h(\mathbf{T}) = 0\} = 1$ for all θ .

A sufficient statistic carries all the information about θ in the data. Once $\mathbf{T}$ is known, the rest of $\mathbf{Y}$ is noise whose distribution does not involve the parameter. Completeness says that $\mathbf{T}$ carries no redundant information either: no nontrivial function of $\mathbf{T}$ has expectation zero for every parameter value. Equivalently, an unbiased estimator that is a function of $\mathbf{T}$ is unique.

Lemma 7.4.1 — Factorization criterion

Suppose $\mathbf{Y}$ has a density $f_\theta(\mathbf{y})$ on $\mathbb{R}^n$. Then $\mathbf{T}(\mathbf{Y})$ is sufficient iff there are functions g_θ and h , with h not depending on θ , such that $f_\theta(\mathbf{y}) = g_\theta(\mathbf{T}(\mathbf{y})) h(\mathbf{y})$ for all $\mathbf{y}$ and θ .

Lemma 7.4.2 — Completeness in exponential families

Suppose $\mathbf{Y}$ has a density of the form

$$f_\eta(\mathbf{y}) = c(\eta) h(\mathbf{y}) \exp\{\eta^\top \mathbf{T}(\mathbf{y})\}, \quad \eta \in \mathcal{E} \subseteq \mathbb{R}^k,$$

with $\mathbf{T}(\mathbf{y}) \in \mathbb{R}^k$. If $\mathcal{E}$ contains a nonempty open subset of $\mathbb{R}^k$, then $\mathbf{T}(\mathbf{Y})$ is complete and sufficient.

Sufficiency follows from Lemma 7.4.1 with $g_{\boldsymbol{\eta}}(\mathbf{T}) = c(\boldsymbol{\eta})e^{\boldsymbol{\eta}^\top \mathbf{T}}$. Completeness rests on the uniqueness of Laplace transforms: $E_{\boldsymbol{\eta}} h(\mathbf{T}) = 0$ on an open set of $\boldsymbol{\eta}$ forces a certain two-sided Laplace transform to vanish on an open set, and hence $h(\mathbf{T}) = 0$ almost surely.

For the normal linear model, expand the exponent of the density:

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}.$$

So the density of Eq. 7.3.1 is

$$f(\mathbf{y}) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}}{2\sigma^2}\right\} \exp\left\{\frac{\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y}}{\sigma^2} - \frac{\mathbf{y}^\top \mathbf{y}}{2\sigma^2}\right\}, \quad (7.4.1)$$

and the data enter only through $\mathbf{X}^\top \mathbf{y}$ and $\mathbf{y}^\top \mathbf{y}$.

Proposition 7.4.3 — A complete sufficient statistic

In the normal linear model Eq. 7.3.1, let $\mathbf{Q}$ be an $n \times r$ matrix whose columns are an orthonormal basis of $\mathcal{C}(\mathbf{X})$. Then $\mathbf{T} = (\mathbf{Q}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$ is complete and sufficient for $(\boldsymbol{\beta}, \sigma^2)$. The statistics $(\mathbf{X}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$ and $(\mathbf{M}\mathbf{Y}, \text{SSE})$ are one-to-one functions of $\mathbf{T}$, so they are complete and sufficient too.

Proof. Since $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X})$, $\mathbf{X}\boldsymbol{\beta} = \mathbf{Q}\boldsymbol{\gamma}$ with $\boldsymbol{\gamma} = \mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta} \in \mathbb{R}^r$, and $\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} = \boldsymbol{\gamma}^\top \mathbf{Q}^\top \mathbf{y}$. By Eq. 7.4.1 the density has the form of Lemma 7.4.2 with $k = r + 1$,

$$\mathbf{T}(\mathbf{y}) = (\mathbf{Q}^\top \mathbf{y}, \mathbf{y}^\top \mathbf{y}), \quad \boldsymbol{\eta} = (\boldsymbol{\gamma}/\sigma^2, -1/(2\sigma^2)).$$

As $\boldsymbol{\beta}$ ranges over $\mathbb{R}^p$, $\boldsymbol{\gamma} = \mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta}$ ranges over all of $\mathbb{R}^r$: the $r \times p$ matrix $\mathbf{Q}^\top \mathbf{X}$ has rank r , because $\mathbf{Q}\mathbf{Q}^\top \mathbf{X} = \mathbf{M}\mathbf{X} = \mathbf{X}$ has rank r . Hence $\mathcal{E} = \mathbb{R}^r \times (-\infty, 0)$, which is open, and Lemma 7.4.2 applies. For the equivalent statistics: $\mathbf{M}\mathbf{Y} = \mathbf{Q}\mathbf{Q}^\top \mathbf{Y}$ (Proposition 6.3.2) and $\text{SSE} = \mathbf{Y}^\top \mathbf{Y} - \|\mathbf{Q}^\top \mathbf{Y}\|^2$ are functions of $\mathbf{T}$, and conversely $\mathbf{Q}^\top \mathbf{Y} = \mathbf{Q}^\top \mathbf{M}\mathbf{Y}$ and $\mathbf{Y}^\top \mathbf{Y} = \text{SSE} + \|\mathbf{M}\mathbf{Y}\|^2$. Also $\mathbf{X}^\top \mathbf{Y} = \mathbf{X}^\top \mathbf{Q}(\mathbf{Q}^\top \mathbf{Y})$, and $\mathbf{Q} = \mathbf{X}\mathbf{K}$ for some $\mathbf{K}$, since the columns of $\mathbf{Q}$ lie in $\mathcal{C}(\mathbf{X})$, so $\mathbf{Q}^\top \mathbf{Y} = \mathbf{K}^\top \mathbf{X}^\top \mathbf{Y}$. A one-to-one function of a complete sufficient statistic is again complete and sufficient, since both definitions refer only to the information (the σ -field) the statistic generates. ■

The reduction is dramatic. An n -dimensional data vector is replaced by $r + 1$ numbers: the fitted values (equivalently, the least squares estimates of the estimable functions) and the residual sum of squares. Under normality the residual *vector*, apart from its length, carries no information about the parameters. Its direction is uniformly distributed over the unit sphere of $\mathcal{C}(\mathbf{X})^\perp$ whatever $(\boldsymbol{\beta}, \sigma^2)$ is (Exercise (C1)). This is the precise sense in which the residuals are “just noise”, and it is also why they are useful for checking the model. Their pattern can reveal departures from normality precisely because, under normality, it has a known and parameter-free distribution.

7.4.2 Rao–Blackwell and Lehmann–Scheffé

Theorem 7.4.4 — Rao–Blackwell

Let $\mathbf{T}$ be sufficient for $\boldsymbol{\theta}$, and let $\delta(\mathbf{Y})$ be an unbiased estimator of $g(\boldsymbol{\theta})$ with $\text{Var}_{\boldsymbol{\theta}} \delta < \infty$ for all $\boldsymbol{\theta}$. Then $\delta^* = E(\delta \mid \mathbf{T})$ is a statistic, it is unbiased for $g(\boldsymbol{\theta})$, and

$$\begin{aligned} \text{Var}_{\boldsymbol{\theta}}(\delta) &= \text{Var}_{\boldsymbol{\theta}}(\delta^*) + E_{\boldsymbol{\theta}}[(\delta - \delta^*)^2] \\ &\geq \text{Var}_{\boldsymbol{\theta}}(\delta^*) \end{aligned}$$

for every $\boldsymbol{\theta}$, with equality iff $\delta = \delta^*$ with probability one.

Proof. The law of $\mathbf{Y}$ given $\mathbf{T}$ does not depend on $\boldsymbol{\theta}$, so neither does $E(\delta(\mathbf{Y}) \mid \mathbf{T})$. It is therefore computable from the data, which makes it a statistic. By the law of total expectation (Proposition 2.5.4), $E \delta^* = E \delta = g(\boldsymbol{\theta})$. By the law of total variance in the same result, $\text{Var} \delta = E\{\text{Var}(\delta \mid \mathbf{T})\} + \text{Var}(\delta^*)$, and $E\{\text{Var}(\delta \mid \mathbf{T})\} = E\{E[(\delta - \delta^*)^2 \mid \mathbf{T}]\} = E(\delta - \delta^*)^2$. This last term is zero iff $\delta = \delta^*$ almost surely. ■

Conditioning on a sufficient statistic averages out the part of an estimator that responds to parameter-free noise. It never hurts, and it helps unless the estimator was already a function of $\mathbf{T}$.

Theorem 7.4.5 — Lehmann–Scheffé

Let $\mathbf{T}$ be complete and sufficient for $\boldsymbol{\theta}$. If $\phi(\mathbf{T})$ is unbiased for $g(\boldsymbol{\theta})$ and has finite variance, then $\phi(\mathbf{T})$ is a **uniformly minimum variance unbiased estimator** (UMVUE) of $g(\boldsymbol{\theta})$: $\text{Var}_{\boldsymbol{\theta}} \phi(\mathbf{T}) \leq \text{Var}_{\boldsymbol{\theta}} \delta$ for every unbiased δ and every $\boldsymbol{\theta}$. It is the only unbiased function of $\mathbf{T}$ and, up to sets of probability zero, the only UMVUE.

Proof. If $\phi_1(\mathbf{T})$ and $\phi_2(\mathbf{T})$ are both unbiased, then $E_{\boldsymbol{\theta}}\{\phi_1(\mathbf{T}) - \phi_2(\mathbf{T})\} = 0$ for all $\boldsymbol{\theta}$, so $\phi_1(\mathbf{T}) = \phi_2(\mathbf{T})$ almost surely by completeness. Now let δ be any unbiased estimator with finite variance. By Theorem 7.4.4, $E(\delta \mid \mathbf{T})$ is an unbiased function of $\mathbf{T}$, so it equals $\phi(\mathbf{T})$ almost surely, and $\text{Var} \phi(\mathbf{T}) \leq \text{Var} \delta$, with equality iff $\delta = \phi(\mathbf{T})$ almost surely. (If δ has infinite variance there is nothing to prove.) ■

7.4.3 Best unbiased estimators in the normal linear model

Theorem 7.4.6 — Minimum variance unbiased estimation

Assume the normal linear model Eq. 7.3.1 with $n > r$, and let $s^2 = \text{SSE}/(n - r)$.

- (a) For every $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is the UMVUE of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$.
- (b) s^2 is the UMVUE of σ^2 .
- (c) More generally, every estimator of the form $\phi(\mathbf{MY}, \text{SSE})$ with finite variance is the UMVUE of its expectation. In particular, for $k = n - r$,

$$b_k = \sqrt{\frac{2}{k}} \frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2})},$$

s/b_k is the UMVUE of σ , and $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + z s/b_k$ is the UMVUE of $\mathbf{x}_0^\top \boldsymbol{\beta} + z\sigma$ for every constant z and every $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$.

Proof. By Proposition 7.4.3, $(\mathbf{MY}, \text{SSE})$ is complete and sufficient, so by Theorem 7.4.5 it suffices to show that each estimator is an unbiased function of it with finite variance. (a) Write $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$. Then $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{MY}$ is a function of $\mathbf{MY}$, it is unbiased by Theorem 7.2.1(a), and its variance is finite. (b) s^2 is a function of SSE, it is unbiased because $E(\text{SSE}) = \sigma^2(n - r)$ (Theorem 2.3.1), and it has finite variance. (c) The first statement is Theorem 7.4.5 again. For σ , note that $\text{SSE}/\sigma^2 = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y} / \sigma^2 \sim \chi^2(k)$ by Theorem 4.3.2(a), since $\mathbf{I} - \mathbf{M}$ is a projection of rank k and $(\mathbf{I} - \mathbf{M}) \mathbf{X} \boldsymbol{\beta} = \mathbf{0}$. If $W \sim \chi^2(k)$, with density $w^{k/2-1} e^{-w/2} / \{2^{k/2} \Gamma(k/2)\}$, then

$$\begin{aligned} E \sqrt{W} &= \frac{1}{2^{k/2} \Gamma(k/2)} \int_0^\infty w^{(k+1)/2-1} e^{-w/2} dw \\ &= \frac{2^{(k+1)/2} \Gamma(\frac{k+1}{2})}{2^{k/2} \Gamma(\frac{k}{2})} = \sqrt{2} \frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2})}. \end{aligned}$$

Hence $E s = \sigma E \sqrt{W/k} = b_k \sigma$, and s/b_k is unbiased for σ , with variance at most $E s^2 / b_k^2 < \infty$. The last estimator is a sum of two unbiased functions of $(\mathbf{MY}, \text{SSE})$. ■

Part (a) is the promised strengthening of Gauss–Markov. Under normality no unbiased estimator of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, linear or not, beats least squares. The midrange of Example (Uniform errors and the midrange) cannot help when the errors are normal. Part (c) shows the method at work beyond linear estimators. The obvious estimate s of σ is biased, because $E s < \sqrt{E s^2}$ by Jensen’s inequality, and dividing by b_k removes the bias.

Example. (Estimating σ without bias)

The constant $b_k = E(s)/\sigma$ depends only on the residual degrees of freedom. For the stack loss regression, $k = 17$ and $b_{17} = 0.9854$, so the UMVUE of σ is $s/b_{17} = 1.0148 s$. The correction is 1.5% here. It is large for very small k , with $b_2 = 0.8862$, and negligible for large k , with $b_{200} = 0.9988$.

```
import numpy as np
from scipy import special

def mean_s_factor(k):
    """b_k = E(s)/sigma when k s^2 / sigma^2 ~ chi^2(k)."""
    return np.sqrt(2 / k) * np.exp(special.gammaln((k + 1) / 2) - special.gammaln(k / 2))
```

```
for k in [2, 5, 10, 17, 50, 200]:
    b = mean_s_factor(k)
    print(f"k = {k:3d}    E(s)/sigma = {b:.4f}    UMVUE of sigma = s / {b:.4f}")
```

[Run this code online](#)

7.4.4 Rao–Blackwellization by reflection

The Rao–Blackwell theorem is not only a device for proofs. In the normal linear model we can carry out the conditioning explicitly, and it takes every linear unbiased estimator to least squares.

Example. (Conditioning on the sufficient statistic)

Let $\mathbf{a}^\top \mathbf{Y}$ be any LUE of an estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, such as the end-point slope of Example (Three linear unbiased estimators of a slope). Let $\mathbf{R} = 2\mathbf{M} - \mathbf{I}$, the reflection that fixes $\mathcal{C}(\mathbf{X})$ and reverses $\mathcal{C}(\mathbf{X})^\perp$. It is symmetric and orthogonal ($\mathbf{R}^2 = \mathbf{I}$) and $\mathbf{R}\mathbf{X} = \mathbf{X}$. Consequently:

- $\mathbf{R}\mathbf{Y} \sim N_n(\mathbf{R}\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{R}\mathbf{R}^\top) = N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ by Theorem 3.3.1, so $\mathbf{R}\mathbf{Y}$ has the same distribution as $\mathbf{Y}$;
- the sufficient statistic is unchanged: $\mathbf{X}^\top \mathbf{R}\mathbf{Y} = \mathbf{X}^\top \mathbf{Y}$ and $\|\mathbf{R}\mathbf{Y}\| = \|\mathbf{Y}\|$.

So the pairs $(\mathbf{a}^\top \mathbf{Y}, \mathbf{T}(\mathbf{Y}))$ and $(\mathbf{a}^\top \mathbf{R}\mathbf{Y}, \mathbf{T}(\mathbf{Y}))$ have the same joint distribution, where $\mathbf{T} = (\mathbf{X}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$. Hence $E(\mathbf{a}^\top \mathbf{Y} \mid \mathbf{T}) = E(\mathbf{a}^\top \mathbf{R}\mathbf{Y} \mid \mathbf{T})$, and averaging,

$$E(\mathbf{a}^\top \mathbf{Y} \mid \mathbf{T}) = E\left(\mathbf{a}^\top \frac{\mathbf{Y} + \mathbf{R}\mathbf{Y}}{2} \mid \mathbf{T}\right) = E(\mathbf{a}^\top \mathbf{M}\mathbf{Y} \mid \mathbf{T}) = \mathbf{a}^\top \mathbf{M}\mathbf{Y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}},$$

because $\mathbf{a}^\top \mathbf{M}\mathbf{Y}$ is a function of $\mathbf{T}$ and $\mathbf{M}\mathbf{a} = \mathbf{a}_*$ (Proposition 7.1.2). The Rao–Blackwell improvement of *every* LUE is the least squares estimate. The listing checks the identity $\frac{1}{2}(\mathbf{a}^\top \mathbf{y} + \mathbf{a}^\top \mathbf{R}\mathbf{y}) = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ for simulated data and the end-point slope.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
M = X @ np.linalg.solve(X.T @ X, X.T)
R = 2 * M - np.eye(n)                                # reflection: keeps C(X), flips C(X)
                                                         )-perp

rng = np.random.default_rng(71)
y = X @ np.array([2.0, 0.5]) + rng.normal(size=n)
y_ref = R @ y
print("X^T y unchanged:", np.allclose(X.T @ y_ref, X.T @ y))
print("y^T y unchanged:", np.isclose(y_ref @ y_ref, y @ y))

a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0]                                # the end-point slope estimator
a_ls = X @ np.linalg.solve(X.T @ X, [0.0, 1.0])
print("average of a^T y over the two points:", (a_end @ y + a_end @ y_ref) / 2)
```

```
print("least squares slope: ", a_ls @ y)
```

[Run this code online](#)

The reflection argument uses the normal distribution only through its invariance under orthogonal maps that fix the mean. It makes concrete what “the residuals are noise” means. Reversing the residual vector changes neither the likelihood nor the sufficient statistic, so an estimator should not depend on the residuals’ direction. An LUE that uses the residuals is replaced by the average of its value at $\mathbf{y}$ and at the reflected data, and that average no longer involves the residuals.

7.4.5 The Cramér–Rao bound

A different route to optimality bounds the variance of every unbiased estimator from below and checks whether the bound is attained. We state the bound for the full-rank normal linear model, where every ingredient can be computed.

Write $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, v)^\top$ with $v = \sigma^2$. The **score** is the gradient of the log-likelihood Eq. 7.3.2, evaluated at the random $\mathbf{Y}$:

$$\mathbf{S} = \begin{pmatrix} \partial \ell / \partial \boldsymbol{\beta} \\ \partial \ell / \partial v \end{pmatrix} = \begin{pmatrix} \mathbf{X}^\top \boldsymbol{\varepsilon} / v \\ -\frac{n}{2v} + \frac{\boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}}{2v^2} \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\boldsymbol{\beta}.$$

Proposition 7.4.7 — Score and information

In Eq. 7.3.1 with $\text{rank}(\mathbf{X}) = p$, $\mathbb{E} \mathbf{S} = \mathbf{0}$ and

$$\mathcal{I}(\boldsymbol{\theta}) = \text{Cov}(\mathbf{S}) = \begin{pmatrix} \mathbf{X}^\top \mathbf{X} / \sigma^2 & \mathbf{0} \\ \mathbf{0}^\top & n / (2\sigma^4) \end{pmatrix}.$$

Proof. $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, v\mathbf{I})$. Hence $\mathbb{E}(\mathbf{X}^\top \boldsymbol{\varepsilon} / v) = \mathbf{0}$ and $\text{Cov}(\mathbf{X}^\top \boldsymbol{\varepsilon} / v) = \mathbf{X}^\top \mathbf{X} / v$ by Theorem 2.2.1. Next, $\boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon} / v \sim \chi^2(n)$ by Theorem 4.3.2(a) with $\mathbf{P} = \mathbf{I}$, so it has mean n and variance $2n$. So the last coordinate has mean $-n/(2v) + nv/(2v^2) = 0$ and variance $2n/(4v^2) = n/(2v^2)$. Finally, by Theorem 4.2.3(c) with $\mathbf{K} = \mathbf{X}^\top$, $\mathbf{A} = \mathbf{I}$ and $\boldsymbol{\mu} = \mathbf{0}$, $\text{Cov}(\mathbf{X}^\top \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}) = \mathbf{0}$. ■

Theorem 7.4.8 — Cramér–Rao bound in the normal linear model

Assume Eq. 7.3.1 with $\text{rank}(\mathbf{X}) = p$. Let $\delta(\mathbf{Y})$ have finite variance and satisfy $\mathbb{E}_{\boldsymbol{\theta}} \delta = g(\boldsymbol{\theta})$ for all $\boldsymbol{\theta}$, with g differentiable. Then

$$\begin{aligned} \text{Var}_{\boldsymbol{\theta}}(\delta) &\geq \nabla g(\boldsymbol{\theta})^\top \mathcal{I}(\boldsymbol{\theta})^{-1} \nabla g(\boldsymbol{\theta}) \\ &= \sigma^2 \mathbf{g}_{\boldsymbol{\beta}}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{g}_{\boldsymbol{\beta}} + \frac{2\sigma^4}{n} g_v^2, \end{aligned}$$

where $\mathbf{g}_\beta = \partial g / \partial \beta$ and $g_v = \partial g / \partial v$.

Proof. Write f_θ for the density. Differentiating $g(\theta) = \int \delta(\mathbf{y}) f_\theta(\mathbf{y}) d\mathbf{y}$ under the integral sign, which is justified because the normal family is an exponential family (Lehmann and Casella 1998, chapter 1), gives

$$\begin{aligned}\nabla g(\theta) &= \int \delta(\mathbf{y}) \nabla \log f_\theta(\mathbf{y}) f_\theta(\mathbf{y}) d\mathbf{y} \\ &= \mathbf{E}(\delta \mathbf{S}) = \text{Cov}(\delta, \mathbf{S}),\end{aligned}$$

the last step because $\mathbf{E} \mathbf{S} = \mathbf{0}$. For any constant vector $\mathbf{c}$, $0 \leq \text{Var}(\delta - \mathbf{c}^\top \mathbf{S}) = \text{Var} \delta - 2\mathbf{c}^\top \nabla g + \mathbf{c}^\top \mathbf{I} \mathbf{c}$. Choosing $\mathbf{c} = \mathbf{I}^{-1} \nabla g$ gives $\text{Var} \delta \geq \nabla g^\top \mathbf{I}^{-1} \nabla g$. The block-diagonal form of Proposition 7.4.7 gives the right-hand side.

■

For $g(\theta) = \boldsymbol{\lambda}^\top \beta$ the bound is $\sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$, exactly the variance of $\boldsymbol{\lambda}^\top \hat{\beta}$. Least squares **attains** the bound, which gives a second proof of Theorem 7.4.6(a) in full rank. For $g = \sigma^2$ the bound is $2\sigma^4/n$, while

$$\text{Var}(s^2) = \frac{\sigma^4}{(n-p)^2} \text{Var}\{\chi^2(n-p)\} = \frac{2\sigma^4}{n-p} > \frac{2\sigma^4}{n}.$$

So s^2 does not attain the bound, although by Theorem 7.4.6(b) no unbiased estimator does better. The bound is simply not sharp here. Its ratio to the attained minimum, the **efficiency** of s^2 , is $(n-p)/n$. That is 0.810 for the stack loss regression, and it tends to one as $n \rightarrow \infty$ with p fixed. The same happens for σ : the UMVUE s/b_k has variance $0.0298 \sigma^2$ when $k = 17$, against the bound $\sigma^2/(2n) = 0.0238 \sigma^2$, an efficiency of 0.798. The equality condition in the proof explains why. The bound is attained iff $\text{Var}(\delta - \mathbf{c}^\top \mathbf{S}) = 0$, that is, iff $\delta = g(\theta) + \nabla g^\top \mathbf{I}^{-1} \mathbf{S}$ with probability one. For $g = \sigma^2$ the right-hand side is $\sigma^2 + (2\sigma^4/n)\{-n/(2\sigma^2) + \boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}/(2\sigma^4)\} = \|\mathbf{Y} - \mathbf{X}\beta\|^2/n$. This is not a statistic, because it involves the unknown β , so no unbiased estimator of σ^2 attains the bound.

Warning. The Cramér–Rao bound is a lower bound, not a target. An estimator that fails to attain it may still be the best unbiased estimator, as s^2 is. When an estimator does attain it, the bound certifies optimality without any appeal to completeness. The Lehmann–Scheffé route is stronger in one respect: it needs no differentiability and covers every estimable function in every rank.

7.4.6 What changes without normality

Every result of this section uses the normal density. Without it the conclusions can fail. Under uniform errors the midrange beats $\bar{Y}$ (Example (Uniform errors and the midrange)). Under Laplace errors the sample median has asymptotic variance $\sigma^2/(2n)$, half that of $\bar{Y}$. Under the second-moment assumptions alone, the best that can be said for s^2 is a result in the spirit of Gauss–Markov. Among quadratic unbiased estimators $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ of σ^2 with $\mathbf{A}$ nonnegative definite, s^2 has the smallest variance when the errors have zero excess kurtosis, or when all leverages are equal. Exercise (C2) outlines this result of Atiqullah (1962).

7.4.7 Exercises

7.4.7.1 A. Check your understanding

Exercise A1.

Assume $\text{rank}(\mathbf{X}) = p$. Use [Lemma 7.4.1](#) and the identity $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \text{SSE} + (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ to show directly that $(\hat{\boldsymbol{\beta}}, \text{SSE})$ is sufficient for $(\boldsymbol{\beta}, \sigma^2)$. Is $\hat{\boldsymbol{\beta}}$ alone sufficient?

Exercise A2.

Suppose $\sigma^2 = \sigma_0^2$ is known. Show that $\mathbf{X}^\top \mathbf{Y}$ alone is complete and sufficient for $\boldsymbol{\beta}$, and that $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is still the UMVUE of every estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$. Why is $\mathbf{X}^\top \mathbf{Y}$ not sufficient when σ^2 is unknown?

7.4.7.2 B. Practice

Exercise B1.

For estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, find the UMVUE of $(\boldsymbol{\lambda}^\top \boldsymbol{\beta})^2$ in the normal linear model. Can it be negative? Comment.

Solution. $E(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})^2 = (\boldsymbol{\lambda}^\top \boldsymbol{\beta})^2 + \sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ and $E s^2 = \sigma^2$, so $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})^2 - s^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ is unbiased. It is a function of $(\mathbf{M}\mathbf{Y}, \text{SSE})$, so it is the UMVUE by [Theorem 7.4.6\(c\)](#). It is negative whenever $|t| < 1$ for the t statistic of [Proposition 7.3.3](#) at $\psi = 0$, which has positive probability. An estimator of a nonnegative quantity that can be negative is clearly improvable, for example by its positive part. The positive part is biased but has smaller mean squared error. Unbiasedness can be a costly requirement.

Exercise B2.

Compute the observed information $-\nabla^2 \ell$ at $(\boldsymbol{\beta}, v)$, take its expectation, and confirm [Proposition 7.4.7](#) by this second route.

Exercise B3.

For the stack loss model, the 90% quantile of the response at a new operating condition $\mathbf{x}_0$ is $\mathbf{x}_0^\top \boldsymbol{\beta} + 1.2816 \sigma$. Write down its UMVUE, and explain why $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + 1.2816 s$ is biased. By how much, in units of σ ?

7.4.7.3 C. Going deeper

Exercise C1.

In the normal linear model with $n - r \geq 2$, let $\mathbf{U} = (\mathbf{I} - \mathbf{M})\mathbf{Y} / \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|$. Show that $\mathbf{U}$ is independent of $(\mathbf{M}\mathbf{Y}, \text{SSE})$, and that its distribution (uniform on the unit sphere

of $\mathcal{C}(\mathbf{X})^\perp$ does not depend on $(\boldsymbol{\beta}, \sigma^2)$. *Hint:* write $(\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{Q}_2\mathbf{Z}$ as in Exercise (C2), and use the rotation invariance of $N(\mathbf{0}, \sigma^2\mathbf{I})$.

Solution. With $\mathbf{Q}_2$ as in Exercise (C2), $(\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{Q}_2\mathbf{Z}$ with $\mathbf{Z} \sim N_{n-r}(\mathbf{0}, \sigma^2\mathbf{I})$, and $\mathbf{Z}$ is independent of $\mathbf{MY}$ by Theorem 3.4.1, since $\text{Cov}(\mathbf{MY}, \mathbf{Z}) = \sigma^2\mathbf{MQ}_2 = \mathbf{0}$. Then $\text{SSE} = \|\mathbf{Z}\|^2$ and $\mathbf{U} = \mathbf{Q}_2\mathbf{Z}/\|\mathbf{Z}\|$. For any orthogonal $\mathbf{H}$, $\mathbf{HZ}$ has the same law as $\mathbf{Z}$, so the law of $\mathbf{Z}/\|\mathbf{Z}\|$ is invariant under all rotations. The only such law on the sphere is the uniform one. The independence of the direction $\mathbf{Z}/\|\mathbf{Z}\|$ and the length $\|\mathbf{Z}\|$ follows from the polar form of the density of $\mathbf{Z}$, which is a function of $\|\mathbf{z}\|$ only. Hence $\mathbf{U}$ is independent of $(\mathbf{MY}, \text{SSE})$, and its law involves neither $\boldsymbol{\beta}$ nor σ^2 .

Exercise C2.

Assume only that the errors are independent with common variance σ^2 , zero third moment and fourth moment μ_4 . Let $\mathbf{Y}^\top\mathbf{AY}$ be an unbiased estimator of σ^2 with $\mathbf{A}$ symmetric nonnegative definite. (a) Show that $\mathbf{AX} = \mathbf{0}$ and $\text{tr } \mathbf{A} = 1$. (b) Using Theorem 2.3.3, show that $\text{Var}(\mathbf{Y}^\top\mathbf{AY}) = (\mu_4 - 3\sigma^4) \sum_i a_{ii}^2 + 2\sigma^4 \text{tr}(\mathbf{A}^2)$. (c) Show that $\text{tr}(\mathbf{A}^2) \geq 1/(n-r)$, with equality iff $\mathbf{A} = (\mathbf{I} - \mathbf{M})/(n-r)$. (d) Conclude that s^2 has the smallest variance in this class when $\mu_4 = 3\sigma^4$. *Hint for (c):* $\mathbf{A} = (\mathbf{I} - \mathbf{M})\mathbf{A}(\mathbf{I} - \mathbf{M})$, so the eigenvalues of $\mathbf{A}$ live on $\mathcal{C}(\mathbf{X})^\perp$.